The future of wireless mesh network in next-generation communication: A perspective overview

Abstract

Wireless mesh network (WMN) which evolves from ad-hoc network is a type of self-healing, self-configuration, and multi-hop wireless network. Without expensive and fixed base stations, WMN can be established fast, easily, and flexibly with low cost. With the fast development of wireless communication, the required data rate and amount increase. As WMN can always provide broad communication access with better scalability, flexibility, and robustness, WMN can still play an important role in future and can bring a wide range of applications in next-generation communication. From the beginning till now, the focus of WMN transfers from sparse connection to intelligent dense networking. The wireless network in future will be more complex with different demands. Considering the features of WMN, this paper provides an overview of WMN from the point of view of next-generation communication. The application scenes including space-air-ground integrated network, Internet of Things, and edge computing are provided. When deploying WMN, the technologies suitable for the demands in next-generation communication, such as blockchain, software defined networking, network functions virtualization, and machine learning are discussed. Further, the upcoming challenges and opportunities in future are also highlighted.

Keywords: Wireless mesh network, Next-generation communication, Internet of things, Edge computing, Machine learning

1 Introduction

Evolving from ad hoc network, Wireless mesh network (WMN) emerges around the mid-1990s and has been extensively studied for nearly two decades. Different from ad hoc network, WMN is more stable and has stronger process ability. WMN is distributed, where nodes can communicate with each other without going through a gateway or base station. There are two types of mesh nodes: mesh routers and mesh clients. Mesh routers have stronger computing ability, more storage space and more radio interfaces than mesh clients. According to the features and components, WMN can be categorized into infrastructure WMN, client WMN and hybrid WMN [1]. Mesh routers and mesh clients constitute infrastructure WMN and client WMN respectively.

Hybrid WMN combines both infrastructure and client WMN. Any type of WMN has the feature of multi-hop communication, which brings flexible coverage and topology. The easy deployment with low cost, high robustness and high capacity makes WMN still a key component in next-generation wireless communication networks [2].

High capacity and high Quality of Service (QoS) are important for WMN at early stage [3] [4] [5]. With fast development of wireless technologies, high efficiency, large coverage and cooperation with low cost become more and more important. To improve the network performance, research including routing, resource allocation, security, and so on has been done. Current research in WMN has many different focuses:

- WMN with other communication technology: WMN can be combined with other technologies like Long Term Evolution (LTE) [6]. LTE can reduce the interference in WMN, and WMN can reduce the cost by buying unlicensed frequency band. A Q learning based routing method is then designed in such hybrid network architecture. The load evaluated by queue length in LTE and the transmission rate in WMN are considered respectively.
- Resource allocation and routing in WMN: Different aspects can be combined together to improve the whole network performance. Routing and channel assignment can be jointly designed to minimize the energy consumption [7]. The mesh nodes are with multiple radio interfaces, and channels will be assigned to the interfaces with mode ON.
- Energy and power control in WMN: Considering energy, joint power control and routing in wireless powered communication networks is researched in [8]. To solve the near-far problem, multi-hop transmission is used to obtain the tradeoff between energy consumption and data transmission request. An infinite horizon Markov decision process model is built to minimize delay. To replenish energy, [9] also considers energy harvesting and proposes a distance and energy aware clustering routing. The node with high energy arrival rate and close distance to the sink will be selected as the cluster head. The cluster head is reassigned periodically to improve the energy efficiency.
- Security in WMN: Security is another important topic in WMN, which can influence the whole network performance. The cross-layer method can be used to obtain link quality in Media Access Control (MAC) layer and the forward action in network layer [10]. The route with selfish node will be avoided, and the path with high reliability will be selected.

There are some reviews concluding the research about WMN. [11] provides a survey about IoT-based WMN, and the wireless communication technologies including IEEE 802.11, Bluetooth, IEEE 802.15.4-oriented, and Sub-GHz-based LoRa are introduced. [12] surveys the status for software-defined WMN. Then some different identified aspects like controlling wireless interfaces, control-plane connection and topology discovery, modulation and coding, routing and load-balancing, client handling are presented. [13] provides a review of the state-of-the-art multihop proposals for LoRaWAN, which is the LoRaWAN mesh network. Technical characteristics, intermediate devices function, and network topologies are analyzed. [14] combines the wireless sensor networks and WMN, and gives a systematic review of existing interconnection

methods. Although existing researches for WMN have covered a wide range of different topics and focuses and there are already some reviews about WMN, the future research of WMN from the point of view of next-generation communication needs further investigation and deserves careful consideration. Therefore, this paper provides an overview about how WMN can be applied in next-generation communication, and there is no such reviews currently to the best of our knowledge. There could be a little misunderstanding that the research about WMN is done in many years and it is not necessary in future. However, as WMN can be flexibly combined with existing network architectures, it is important with a great application and expansion potential in future. The WMN should adapt to new scenes and applications by using state-of-art technologies. Given that the data transmission rate and user amount will increase dramatically in next-generation communication, it will become more and more important to deploy WMN effectively and improve network resource efficiency.

For the above reasons, this overview will review the application potentials of WMN, the further technologies and research directions from a perspective of next-generation communication. Keep such a goal in mind, the rest of the paper is organized as follows: starting from the analysis the applications of WMN in future, an overview of the possible new designs and technologies is then followed. By combining the needs of the future applications and the possible new technologies, the future challenges and research directions are analyzed and identified. Finally, a summary of the main views is provided at the conclusion section.

2 Applications of WMN in future

Based on the characteristics of WMN, currently it is often deployed in military communication, emergency communication, community communication, and so on. However, with the fast development of wireless communication, new requests and scenes are appearing. Therefore, the role and application scenes of WMN in future need to be fully recognized and carefully considered. Since the research about WMN has been done for several decades, and it seems the WMN is not a new technology and may be obsolete by era. However, WMN can adapt to the development of wireless communication, and the next-generation communication brings new application scenes and challenges for WMN.

Although the services of Internet are well provided in well-developed economies and 5th generation mobile networks (5G) start to be established commercially, there are still many regions and areas lacking connectivity nowadays. For example, some rural areas far away from cities have precipitous landform and residents are scattered. It is very difficult and expensive to build base stations of cellular networks in such areas even today. The use efficiency of base stations is low because of the sparse habitations. There will be only a few users covered by one base station if built. The capability and resource of the base station are then wasted. WMN can play an important role in this case just as it does now. Companies like SmartMesh [15], RightMesh [16], Ammbr [17], and BlockMesh [18] start to commit to accelerate the deployment of connectivity via WMN nowadays. In addition, WMN can still be used in some new scenes like satellite-terrestrial integrated networks and unmanned aerial vehicle (UAV) communication

networks. Based on the recent research trends and the relevance in the industry, some applications of WMN in future are selected and given in this section.

2.1 Space-air-ground integrated network

In last few years, big data, smart industry, 5G, 6G and so on develops, which causes the explosive increase of users and service requests on ground. Next-generation communication needs to provide more resource to satisfy the large amount of service requests. With the limitation of communication coverage and capacity, only wireless network on ground cannot implement the reliable and fast data transmission globally. To solve such problem, the space-air-ground integrated network (SAGIN) emerges [19]. SAGIN can be used in earth observation and mapping, intelligent transportation, military missions, disaster relief, and so on. The requests in SAGIN are usually flexible and needs to be served with high reliability, which matches the features of WMN. Therefore, WMN and its extended researches can be applied in SAGIN.

SAGIN is still at the beginning stage, and it has multiple layers including space layer, air network, and ground network [20]. To transmit data flexibly, WMN technology can be combined in these layers. In space layer, the coverage is large but the propagation latency is long. In air network, the coverage is wide, cost is low, and deployment is flexible but the capacity is less, link is unstable and mobility is high. For ground network, the delay is very low, resources are rich but the coverage is limited. SAGIN therefore combines the advantages of these layers together to serve different types of requests. The nodes in SAGIN actually need to connect with each other and exchange data. Such network architecture is suitable for WMN. With the help of WMN, the topology will be flexible to suit the mobility of nodes like UAV, and the technologies like load balance can be applied in SAGIN. The resource orchestration problem is essential in SAGIN [21]. To solve the problem of the resource limitation in SAGIN, the resource allocation and optimization technologies in WMN can also be used.

In some particular cases like oil well detection at sea, UAV and satellite communication play an important role. In this case, traditional communication technology based on base stations is not available because of the environment on the sea. UAV will therefore support to transmit the collected data. In some large areas, only one UAV cannot satisfy the detection requests. Several UAVs need to cooperate to detect the area. They need to communicate with each other to confirm the checked area and information. Then they can design the best detection path and computation offloading together to cover the whole area with least cost [22]. The UAVs can fly periodically to guarantee the smooth work of oil wells. If something wrong is detected like solder joint falling off, the UAV can report the problem as quickly as possible to minimize the loss. During the process of UAVs cooperation and data exchange, this scene fits WMN perfectly. The UAVs can build a network in self-organization way to flexibly exchange data without fixed base stations. Therefore, WMN can play a role in this case.

From the general layers of SAGIN, we can see the mobility is getting higher from the top layer (e.g. space layer) to bottom layer (e.g. ground network). This feature is similar to the hybrid WMN. The nodes in higher layer can be seen as the infrastructure

nodes, and those in lower layer can be set as mesh clients. Therefore, WMN can be applied in SAGIN in next-generation communication very well.

2.2 Internet of Things

Internet of Things (IoT) develops with a very fast speed. There are more than 40 billion connected IoT devices in 2020, and will be more than 125 billion by 2030 [11]. This is a very large number, and it will increase further. The things in our daily life like lights, cleansing robots, sensors, doors, bikes, and so on can be connected. It is important to design such a network effectively, as it will greatly enhance the connections and significantly improve the user experience.

In short-range IoT communication, Bluetooth, Wi-Fi and ZigBee are widely used [23], and WMN is a key technology-agnostic enabler [24]. The coverage area can be extended by multiple hops in WMN. Mesh networking has already been supported by Wi-Fi and ZigBee, and some products can be found on the market. Wi-Fi can provide high throughput, but the energy consumption is also high. ZigBee is widely used, but it needs extra hardware. As Bluetooth has been equipped on smartphones, tablets, TV, and some other electronics, it can be used more conveniently and easily without the request of particular hardware. Besides, when consuming similar energy, Bluetooth can provide higher data rate [25]. However, the extensibility and short coverage are the main disadvantages of traditional Bluetooth networks. Mesh topology can solve this problem by enabling relaying and many-to-many communication. Bluetooth mesh is officially launched in 2017, and the mesh networking still needs support. Companies like Cisco and Ericsson have led some related research. The scenes that IoT-based WMN can be deployed include monitor forest fire, package delivery, underwater tube detection, UAV communication, and so on. Although the technology of IoT Mesh is yet to be standardized, it has the great potential in terms of power efficiency, connectivity, and confidentiality [26]. The types of network that can compose IoT-based WMN and the applications of IoT-based WMN are given in Fig. 1.

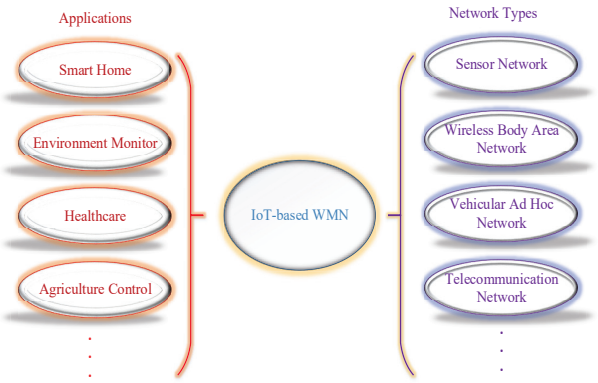


Fig. 1 Applications and network types of IoT-based WMN.

Some studies about IoT-based WMN start to emerge. In IoT-based WMN, some real platforms [27] like radio mesh platforms have been built to evaluate ideas of power control [28], hybrid IPv4/IPv6 dual stack environment [29], exchange technology [30], wireless communication protocol [31], topology design [32], coverage expansion [33], distinguishing abnormal nodes [34], testing network performance [35] [27], improving lifetime with data rate adaptation and information storage [36], energy saving in lighting system [37] and edge computing (EC) [38]. At the same time, field experiments have attracted the researchers' attentions [39]. However, all these existing platforms are not large, and the network topologies are relatively simple. To be closer to reality, large-scale platforms are still needed to be developed in future.

Some other researches focus on the theoretical design. Main research fields include routing [40] [41] [42], resource allocation [43] [44], optimizing lifetime [45], topology control [23], protocol challenge analysis [46], performance analysis [47] and security [48]. When doing the network design, several optimization models and technologies such as integer linear programming [40], integer nonlinear programming [45], genetic algorithm [32], reinforcement learning (RL) [49] and game theory [44] can be used to minimize a cost function with some constraints including energy, delay, packet loss, hop count, geographical location and throughput. The cost function is designed according to the application request. To better manage data traffic, reduce routing overhead and improve scalability, clustering can also be designed [50] [51]. Selecting cluster head and control packet flooding is important. Network conditions may change all the time, so replanning when the network performance decrease is detected or periodically can adapt to the dynamic network conditions well [52]. In security, public-private key cryptography can be used to protect authentic data.

Different research aspects are not independent, and they should be designed jointly. For example, a more reliable routing will be obtained if considering mobility [53], energy [54] [55] [49], spectrum [56], channel allocation [57] and avoiding attacks [58] [59] at the same time. Before routing, the topology formation can be done to provide the selection space [60] [61]. To improve the energy efficiency and reliability, the number of routes, gateways and lifetime can be considered together [45]. These aspects can influence each other, so it is better if consider them jointly.

2.3 Edge computing

With the fast development of IoT, EC has become an important research area to efficiently allocate the computing resource and reduce the burden of cloud. Cloud computing has the strong capability of computing, but with long delay and low mobility. Gartner defines EC to solve data processing at or near the data sources. Thus, the trend from centralized to distributed computing appears. EC serves as the decentralized extension of the campus networks, cellular networks, data center networks or the cloud [62]. Ultra-dense IoT mesh networks and machine-to-machine communications herald an enormous opportunity for new computing and communication paradigms.

As a new Gartner report says, around 10% of enterprise-generated data is created and processed outside a traditional centralized data center or cloud. By 2025, Gartner predicts this figure will reach 75% [63]. The application use cases include augmented and virtual reality, manufacturing, video monitoring, healthcare, smart home,

emergency building evacuation, and automotive vehicle. Cloud, fog and cooperative computing are three main computing paradigms in IoT. The trend of the paradigms is from centralized to distributed. The demand of distributed computing increases because of the short-delay and flexible request. In this case, the distributed feature of WMN can work well, and WMN can be deployed to bring a new paradigm to enable distributed intelligence [64]. The structure of applying WMN in EC is shown in Fig. 2.

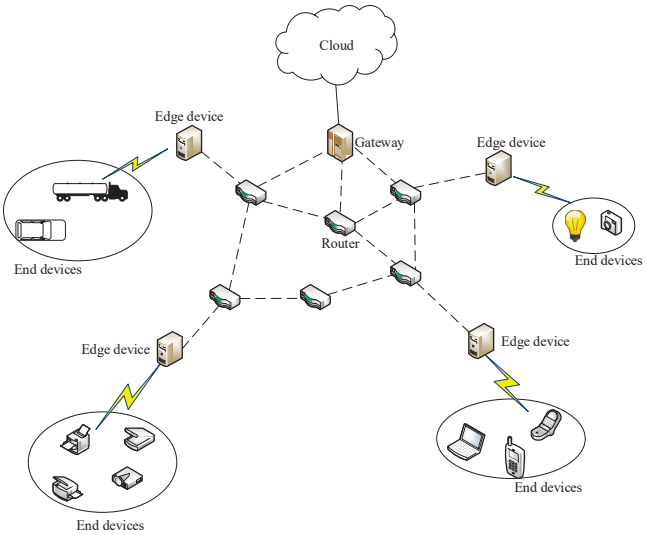


Fig. 2 WMN in EC.

As fully distributed paradigm brings high signal overhead, integrating different computing paradigms is essential. The implementation of WMN in distributed intelligence can achieve this because of the distributed features in WMN. Different types of end devices will be integrated and can cooperate and interact through edge devices. The whole network is hierarchical and edge devices compose the WMN as shown in Fig. 2. Routers relay data packets among edge devices. The edge devices can be a smartphone which is accessed by end devices like lights, cameras, etc. The end devices take responsibility of sensing and collecting data. The data do not need to be all sent to the centralized server, because edge devices also have the ability of computing and making decisions. The edge devices can be responsible for the scheduling of the data, converting data into required format, and making decisions for the resource allocation. The processing and reply time can then be saved. All data and computation tasks are shared using the WMN of edge devices and routers. When allocating tasks, data distribution, device constraints and heterogeneity should be considered.

With the benefit of WMN, the fault tolerance is improved. Some alternative connections among edge devices are provided to better guarantee task computation. As more than one edge devices exist, tasks can be divided and distributed to different edge devices to solve the bottleneck problem. In addition to the traditional connection between a device and router in WMN, implementing WMN in EC can also enable

the interaction between end devices and cloud. The edge devices can not only forward data packets, but also decide how to process, store computation tasks and share data.

We can see that WMN can play a critical role and become an important research area with the development of IoT and EC [65]. At the same time, to realize the great application potentials of WMN in future, some new and up-to-date technologies are required.

3 Technologies in WMN design

To support the applications of WMN in next-generation communication mentioned above, some intelligent technologies are needed to be combined in network design. In general, existing technologies in WMN can be classified in different dimensions:

- Data transmission and processing technologies. 1) High-speed data transmission technologies, including: 5G/6G; Millimeter-wave communication; Multiple input multiple output technologies. 2) Data processing technologies like applying computing and storage resources at network edge for data processing and analysis, which can reduce latency.
- Network management and optimization technologies. 1) Self-organizing network technologies, including: Automatic configuration, management, and optimization of network settings; Self-healing and automatic fault recovery. 2) Artificial Intelligence (AI)-driven network management such as machine learning (ML) based methods for network traffic management, fault prediction, and resource allocation.
- Energy efficiency and sustainability technologies. 1) Low-power network technologies, including: Energy-efficient hardware; Energy-saving transmission and network protocols. 2) Renewable energy integration and energy harvesting, which can integrate renewable energy technologies like solar and wind energy to achieve green networking.
- Security and privacy protection technologies. 1) Encryption and security protocol technologies, including: Advanced encryption standards; Secure authentication; Key management mechanisms. 2) Privacy protection technologies, including: Anonymous communication; User-controlled privacy settings.

WMN in future will be much more dense, complicated, and with high interference. Based on the existing technologies, technologies adapted to future WMN are further needed. For the security and privacy protection, distributed authentication and authorization mechanisms need to be further achieved in future. Blockchain technology is effective to manage the network access. As the topology of WMN in future is very complex and hardware resources are limited, software control can improve resource utility and manage network efficiently. Software defined networking (SDN) and network functions virtualization (NFV) are therefore important for data processing and network management in future. Although the WMN in future will process huge amounts of data, the data needs to be processed in real time, with very low delay and energy consumption. The network should also be self-adaptive and give strategies intelligently according to the dynamic network conditions. In this case, ML can work

well. There are already some ML-based methods nowadays, but they should be further designed according to future applications and requests. Therefore, the methods including blockchain, SDN and NFV, and ML are presented as follows.

3.1 Blockchain

Blockchain is a distributed ledger that is consensually shared, trusted and synchronized. Unlike centralized repositories, blockchain can hold transactions and records in a decentralized way. No centralized authorities are needed. As WMN is a distributed network, distributed ledger can operate well in WMN.

Motivated by blockchain tokens, when combining the blockchain with the WMN, people themselves have the right to connect to each other. The challenge to scale of WMN in last-mile Internet access and SAGIN can be solved by economic incentives. In this way, users can buy or sell communication resource like capacity and transmission rate. User nodes can pay their neighboring nodes to send out data. The carrier nodes which are paid can forward data packets to others. The users with good network resource are encouraged to share their resources with others, and the sustainability of the WMN business models is increased. With the point-to-point messages, the risk of exposing information is reduced. The securely distributed ledgers that form the basis of the blockchain make it possible for applications working without a trusted central node. Thus, establishing large-scale WMN to cover the countries with poor network connections is feasible and valuable. Besides, in cloud and EC, blockchain can provide efficient and incentive mechanism and trust to motivate the edge and micro cloud resource providers to join and constitute a more generalized and decentralized ecosystem [66]. The resources can then be shared to offer seamless service, which will reduce the difficulty of running distributed applications on heterogeneous resources. The use of smart contracts enables the automated negotiations between different operators to help provide high-quality services together [67].

In addition, as the concept of blockchain has been extended to broader contexts, a transaction is no longer necessarily related to the exchange of crypto currency. Data or other values needing to be validated can also be a transaction [68]. Blockchain can be used to manage trust between the nodes and improve security and privacy. Anonymous access control and secure authentication can be implemented. As each node reserves a copy of the entire chain and has visibility of other nodes composing it, the node with local network information can be verified and communicate with others. The network status is more visible. Resource allocation can also use blockchain technology. The resource owners can be protected from the double spending attack because a transaction is validated only it is consistent with the stored ledger in most blockchain nodes. The blockchain consensus mechanism can also be used for edge computing-enabled IoT systems to prevent malicious nodes from participating in consensus [69]. Although the blockchain is famous and effective for its security as a newly emerged distributed framework, the time efficiency to achieve consensus is one of the bottlenecks. This shortcoming should be paid special attention when applying blockchain in applications.

3.2 Software defined networking and network function virtualization

The emerging SDN is an important technology in 5G, which decouples the control and data planes. SDN can improve network management and allow the programmability of control plane [70]. The complex logic functions do not need to be installed in the switches and routers, and the rules which are already computed by the centralized controller will be installed to the switches. Switches on a data plane can forward data simply according to the rules. With SDN, organizations can better achieve edge technologies in EC by flexibly defining rules on where and how data are processed [71]. Application performance and user experience are then be improved. Besides, SDN makes it possible to develop a multi-layer approach to security that takes the communication layer, hardware layer and cloud security into consideration simultaneously [62].

To reduce the difficulty of managing WMN, SDN is an effective technology to overcome the physical limitation [2]. In the mentioned applications of IoT and EC-based WMN, there are a large amount of devices. It will be difficult to control and coordinate them. The communication cost among them will be high in traditional WMN. With the help of SDN, the controller can know the topology and link information of the whole network, so the cost of exchanging information between nodes can be saved. Openflow is an open communication protocol, and it can be the basis of network design. It provides the communication channel between the control and data planes and serves as the building block of SDN solutions.

To achieve data transferring in network like vehicular network, different types of wireless links can be used among vehicles and between vehicles and base stations. If no valid path exists, vehicles will send request to routing server [72]. After getting the rules from the controller, the communication between vehicles can start. Besides Wimax, cellular networks can provide the communication between nodes and controller [73] [74]. The QoS and cost should be considered to improve routing effectiveness [70]. To balance load, multiple gateways can provide service simultaneously and a new path will be switched if tested performance drops [75].

In resource optimization, when assign wireless resource like channels, the channel switching latency should be analyzed [76]. The channels with less congestion and better performance can then be switched. The efficient spectrum allocation and scheduling algorithms which satisfy both control and data traffic can improve network throughput [2]. The access point selection, gateway selection and routing can be considered together [73].

Some hybrid data forwarding methods combining traditional routing methods and SDN can also be designed, which will enable SDN to be easily compatible with current network deployment [77]. In this case, when SDN devices, gateways or controllers have any issues, the traditional nodes can guarantee communication [60] [78]. Moreover, to balance load, when the controller has heavy load, only the most important data flows will be processed by the centralized controller. The less important data flows can be processed by alternative original distributed mechanism [79].

SDN can provide more flexible possibilities for WMN [74] [79]. Better mobility management, routing and resource allocation can be achieved. The combination of

SDN and WMN has better performance than pure WMN [78]. Service requests can be satisfied by simply updating software rather than changing hardware. Applying SDN to traditional WMN still needs to be investigated further in order to take full advantages of both SDN and WMN. Global information can be obtained by the support of SDN, but the cost of uploading local information to the network manager should be carefully considered. The resource allocation to transmit control and data packets needs to be designed efficiently. Besides, the application of SDN in WMN is still at the early stage in reality, so real platform should be further implemented to test and evaluate network performance.

Similar to SDN, NFV can also improve flexibility, reduce cost, and support scalability with the help of software control. NFV is a concept about the network architecture. Different from the traditional network devices (e.g., routers, firewall, load balance, etc.) having their own hardware and software systems, NFV learns from the structure of x86 server and encapsulate such network functions into modular software. When different modular softwares are run on hardware devices, different network functions can be achieved on single hardware device. Software and hardware are no longer tightly bonded. Similarly with SDN, NFV also sets the network virtualization as the objective and improve the network management efficiency. But NFV relies on the standard hardware device.

To satisfy the quality of experience demands of IoT applications, NFV often combines with EC. The IoT service can be expressed as a service function chain (SFC), which is composed of several strictly ordered virtual network functions (VNF) [80]. As mentioned in EC, WMN can play an important role. Therefore, WMN features can also be considered in VNF placement problem to minimize the end-to-end delay and maximize network performance. The NFV technology can be naturally applied in the EC-enabled WMN. The problem of VNF placement can then be combined with computation offloading [81], resource allocation [82], and routing [83]. The routing design in NFV can be further studied because current research usually use the path with least hops. Although NFV makes the network deployment more flexible and improves the resource efficiency, it may cause a little high complexity.

In general, SDN and NFV can be used together to achieve complete network virtualization. They have complementary functions. SDN provides flexible control over network traffic by centralized management, while NFV focuses on the network services virtualization to improve the hardware utility efficiency. When combining them together, SDN can provide efficient routing of data traffic, and NFV can deploy the corresponding network services. The flexibility, ability of programming and cost efficiency can be improved by using them together.

3.3 Machine learning

Although the research of WMN is not new and WMN has been extensively studied already, the fast development of wireless communication brings more challenging requests in recent years. The consumer demand for WMN has only recently caught up, rendering efforts at optimizing WMN to support high capacities and offer high QoS, while being secure and fault-tolerant, more important than ever. When 5G rolls out, massive amount of devices and data appear in mentioned application scenes of

WMN like user access connections, SAGIN, IoT and EC. The networking topology, user requests, and computing tasks are much more complicated than before. Due to the complex conditions and potential states, it will be difficult for human to give best solutions according to the known knowledge. More intelligent methods are needed to solve the network design problems in WMN. As ML is a more and more heated technology which can solve many complicated problems nowadays, there is a trend to apply ML to network design in WMN [84]. ML is very effective to get near-optimal solutions quickly in problems with high dimensionality, which can process data and tasks efficiently and intelligently [85]. Unlike traditional rule-based approaches, ML is more robust and adaptive for changing network environment, which can implement real-time optimization. Knowledge and information can be learnt from the data automatically without so much influence from rules and human behaviors. Network performance can then be improved by analyzing the existing collected data.

ML is categorized into supervised, unsupervised and RL. These ML technologies can be used to solve different problems. Supervised learning method learns through training sets with known optimal solutions, which can bring good performance. Low end-to-end delay, tree cost, computational time, load and high network throughput can be achieved together [86]. However, the process complexity is high. Imbalanced input data may cause errors and overfitting [87]. Unsupervised learning tries and evaluates the performance by itself. The particular unsupervised learning method K-Means can be used to distinguish good and malicious nodes according to reputation values [59]. Security is then improved. Besides, in node clustering, when identify nodes' behavior like location and mobility to build layered network, the complexity of network maintenance can be reduced. K-Medoids which is a variation of K-Means is less influenced by extreme values than K-Means [32]. Nodes can be deployed to extend coverage area and build more robust topology.

It is difficult to model the complex network condition in advance. Thus, RL is popular in network design because no specific model is needed, which is beneficial to making decisions according to the rewards and interaction from environment [88]. Q-learning is one of the most widely used RL algorithms. To define reward function, energy [89], distance [88], lifetime [90], location [91] [92], queue length, transmission rate [6] [93], bandwidth, delay, packet loss [87] [94] can be considered based on different design goals. The reward function can be learnt with the evolving of the network environment [95]. As many nodes have energy limitation, energy harvesting is also an important factor with increasing attention [96] [97]. Energy can be saved by limiting the transmission rate of control packets [98] [99]. When balance exploitation and exploration, more packets can be transmitted along the path with the best quality and then less packets are forwarded along other routes [100]. When the agents interact with environment, the transition model can be generally learnt and estimated online to increase the convergence speed [96]. Besides Q-learning, learning automata is another effective approach. Different from Q-learning, the probability of all actions will be affected once an action is selected. Distributed decision making can be done by learning automata in highly stochastic environments. In a distributed network like WMN, each node knows local network information, so cooperation between nodes can help to find the global optimal solution [87]. In addition to single-objective problems,

RL can also solve multi-objective problems without the assumption of convex [101]. Further, RL can combine with SDN, by utilizing the benefit of programmability and capability of intelligent routing of SDN [102]. The adaptation of ML to solve network design problems is still at its primary stage but with great potentials [87]. Therefore, it needs to be further researched and developed in future.

Deep learning (DL) is a new and very active research area in ML, which is closer to the objective of AI. It has achieved many successes in areas of computer vision, voice recognition, and natural language processing. With the benefits of automation, DL can also be used in WMN design. Using DL to control data flows is a new area. The congestion, delay, packet loss rate and so on will be learnt to avoid errors that occurred before, and the data flows can be controlled in real time. Features can be extracted and learnt from the network traffic. Traditional routing protocols may even be removed and replaced by DL technology in future [103]. DL can also be used to predict link, load, and congestion condition which are the foundation of resource allocation. It needs a large dataset which represents all the possible scenarios to train the neural network [49]. When predict network condition, online learning can better adapt to the changes with high accuracy [104]. Similar to RL, DL can also be combined with SDN technology [105]. With the help of SDN, software defined routers can run deep belief architecture to obtain next hops [106]. Considering the topology feature of WMN, graph-based DL can achieve good performance. Further, the graph convolutional and graph attention networks in graph-based DL models work effectively in SDN [107]. Graphics processing unit with highly parallel computing capability becomes the most viable hardware platform to run DL algorithms, which can accelerate the running speed effectively. Agents can adjust automatically based on the traffic conditions. Deep RL is also worth to be tried and tested for its effectiveness in traffic scheduling in real network [108].

Using ML in network design of WMN can achieve better performance and satisfy flexible service request. However, the research about taking advantage of ML in WMN is still a long way to go and substantial further researches are needed. The time cost and hardware request of running ML should also be further considered to check whether ML is suitable for current application requests.

According to the different requests and objectives, all the technologies mentioned in this section can be used to support the applications of WMN in next-generation communication. In summary, the features, advantages, and weaknesses of the mentioned technologies are given in Table 1.

The applications and research of WMN in future and the available technologies are concluded in Fig. 3.

The application scenes of WMN in next-generation communication include SAGIN, IoT, and EC. These scenes can be considered and addressed together. With different application scenes and application requests, there are many research aspects to be solved. The research aspects in deploying WMN cover power control, topology design, prolonging lifetime, routing, resource allocation, security, performance analysis, etc. They are not fully independent, and can influence each other. For example, different transmission power will bring different transmission range, different neighbour and interference relationships. The topology and network lifetime will also change. To

Table 1 Comparison of different technologies

Technologies	Features	Advantages	Weaknesses	Usage
Blockchain	Distributed ledger to have incentive mechanism for users to join and share	Decentralized, open, independent and secure	Low time efficiency	All these technologies can be used alone or cooperatively in the applications of WMN in future according to the network demands and features
SDN	Decoupled control and data planes	Improve network management and allow programmability, high flexibility	Cost of uploading local information may be a little high	
NFV	Virtualize the traditional network functions into software to decouple software functions and hardware	Several functions can be run on a single hardware server, which can bring flexible services, less spending, and greater resource efficiency	High complexity on reliability design and implementation	
ML	Network has the ability to learn and optimize, which can process data and make policies intelligently	Be suitable for dynamic network features and can make decisions in real time	Need some computing capabilities and some time to train	

improve the whole network performance, methods of routing and resource allocation will be influenced according to the changed topology. Similarly, when changing any aspect, network conditions will change, which will affect the design of other aspects. Therefore, different aspects can be considered at the same time based on different QoS requests. These problems are more complicated because of the large amount of data and dense users in WMN of next-generation communication. When solving these detailed research aspects with multiple objectives, intelligent technologies like blockchain, SDN and ML can be utilized and applied. The combination of these technologies is also very effective to solve different parts of a problem.

4 Challenges and opportunities

Although WMN has many potentials in next-generation communication, there are many challenges which need to be addressed when implement the network design and materialize these potentials.

4.1 Challenges

4.1.1 Node heterogeneity

In next-generation communication, the network scene is complex and may include many different types of nodes like SAGIN. A WMN may be composed of different types of nodes supporting different wireless technologies. New air interface, new applications, new scenes and new architectures exist in next-generation communication. In such distributed and heterogeneous network systems, cooperation and synchronization are very important and difficult. Communication among nodes using different communication protocols should be standardized [64]. The wave band like Ku, Ka, K,

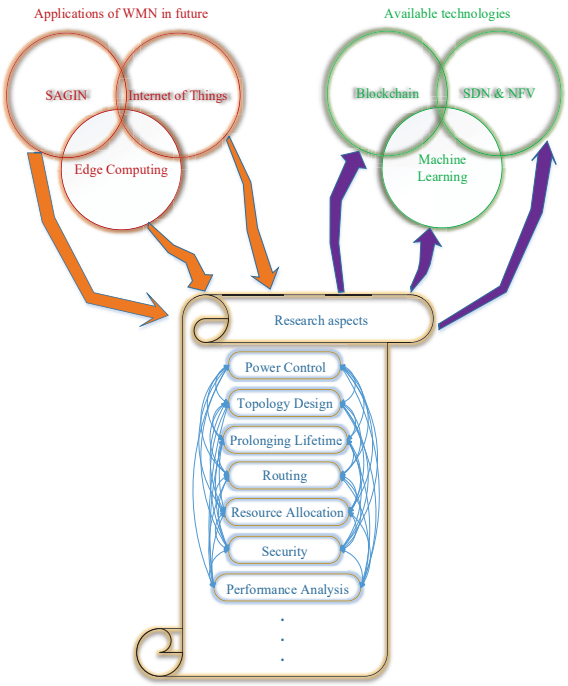


Fig. 3 Overview of applications, available technologies and research aspects in WMN.

etc. in space are easily affected by rain and atmospheric attenuation. Besides, different devices have different processing abilities. How to coordinate them to support the demand of WMN with low cost is challenging. In EC, process resources may be provided by different service providers, so standard, compatibility and interoperability are needed [11]. In industry, in addition to general servers like CPU (Central Processing Unit) and GPU (Graphic Processing Unit), FPGA (Field Programmable Gate Array), and RDSP (Reconfigurable Digital Signal Processing) can also provide computation functions. How to select different computing resources based on different duties with different goals and requests should be carefully considered. The information of the available computing resources should be shared, and the tasks will be allocated to proper resource, which can satisfy the QoS requests. The computing time and network performance can then be improved. An incentive mechanism among particular resources should also be proposed and designed to encourage resources to evenly participate in processing tasks. In this case, SDN and ML technologies can be applied to adaptively allocate the network resources based on the network conditions.

4.1.2 Multi-source heterogeneous data

As there are different types of nodes in next-generation communication, the data generated by them is also different and heterogeneous. We should not only make the different nodes cooperate and allocate resource among them, but also the multi-source heterogeneous data should be processed properly. In the same industry IoT-enable

WMN, heterogeneous data may be collected at the same time. For example, the intelligent photoelectric pod systems in next-generation industry IoT may collect different types of data. Cameras for iron tower, unmanned surface vessel, and reconnaissance plane will collect different images because their different source devices, objectives and environment. Radar images, infrared image, and visible image can be obtained. To process these images and obtain the 3D fusion image, the features of these heterogeneous data need to be extracted and fused. In this case, ML technology can solve this problem well. These different images can be input to multi-modal deep neural networks, so that the features will be fused. The data cleaning of original data can be done, and the incomplete data and noise will be deleted. The obtained data elements can build the final fusion image.

4.1.3 Security

Security is one of the important aspects in network design. With the fast development of wireless communication, more and more information and data are included and transferred nowadays. In IoT-based WMN of the next-generation communication, more types of information including temperature, using time, electricity consumption and so on will be forwarded and analyzed. It will be more dangerous than ever if these information is exposed, because the packets transferred among devices in next-generation communication contains more detailed and private information. In network management and policy making process, to avoid transmitting private data, the federated learning and transfer learning in ML can be applied. In such technologies, the data of users do not need to be sent out but only exchanging the learnt models. In this way, the security can be guaranteed and the network management can make decisions from a global way. The work in this area needs to be further researched.

In addition, the wireless multi-hop nature and ad hoc connections among users of WMN expose the network to many security vulnerabilities [109]. The attacks will decrease network performance and waste wireless resources, so the manipulation, deletion and packet loss need to be avoided. The confidentiality, authorization, anonymity, and fairness should be guaranteed. In next-generation communication, the network environment of WMN is very fast-changing, so the intrusion and authorization need to be designed to adapt to the frequent joining and leaving of nodes.

4.2 Opportunities

1. For the challenge of node heterogeneity

   Considering the challenge of node heterogeneity, the opportunities include:

- Improved QoS and experience. Different types of nodes can provide customized service according to users' requests. For example, some nodes can provide high-rate data transmission and some nodes can achieve long-distance connection. Some new applications and services can be supported.
- Flexible network deployment and management. Heterogeneous networks can provide flexibility and satisfy different application scenes. The technologies of dynamically adjusting resource allocation can be improved.

- Improved energy efficiency and reducing cost. As the heterogeneous nodes have different features and advantages. When allocating the communication tasks to the proper type of nodes, the whole network performance can be improved.

2. For the challenge of multi-source heterogeneous data

Considering the challenge of multi-source heterogeneous data, the opportunities include:

- Data-driven intelligent strategy. By analyzing the multi-source heterogeneous data, the needs and behaviour of users can be observed more clearly. More accurate and intelligent strategies can then be made to provide customized service.
- Improved security and privacy protection. Security threats can be found timely by analyzing the multi-source heterogeneous data for different sources separately. The data privacy can then be protected.
- Improved Automation. IoT and industry in future will be based on multi-source heterogeneous data. The effective management and analysis of such data can improve automation level in IoT.

3. For the challenge of security

Considering the challenge of security, the opportunities include:

- The innovation of security technologies. To deal with the advanced security threats, more sophisticated security technologies and protocols will be developed, which includes the cryptography, intrusion detection, network monitoring, and security management. The innovative technologies will improve the network reliability, and WMN can defense more attacks to guarantee QoS.
- Improved cooperation and standard setting. Different users and groups meet different security threats. This can promote regulatory cooperation and the establishment of security standards. The security problem will be solved more efficiently in future.

In short, there are still many challenges when deploying WMN. However, the opportunities are always together with challenges. The performance improvements and potentials of WMN can be further materialized if these challenges can be solved.

5 Conclusions

With the features of low cost, flexibility, easy deployment, and high robust, WMN still has a key role in next-generation communication. Coverage range will be extended by multiple hops. Due to the flexibility, distributed features and expanded coverage, WMN is adaptive in SAGIN, IoT and EC in future. The developing technologies like blockchain, SDN, NFV and ML are beneficial to improving the whole network performance in different application cases. Different technologies can be chosen according to the requests, objectives, and network conditions. As WMN is a distributed multi-hop wireless network, when applying WMN in next-generation communication, there are still many challenges needing to be solved with the upcoming applications. The challenges like node heterogeneity, multi-source heterogeneous data, and security should be further researched and studied.

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